

OpenPointCloud: An Open-Source Algorithm Library of Deep Learning Based Point Cloud Compression

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ABSTRACT

This paper gives an overview of OpenPointCloud, the first open-source algorithm library containing outstanding deep learning methods on point cloud compression (PCC). We provide an introduction of our implementations, including 8 methods on lossless geometry PCC and lossy geometry PCC. Principles and contributions of these methods in our algorithm library are illustrated, which are also implemented with different deep learning programming frameworks, such as TensorFlow, Pytorch and TensorLayer. In order to systematically evaluate the performances of all these methods, we conduct a comprehensive benchmarking test. We provide analyses and comparisons of their performances according to their categories and draw constructive conclusions. This algorithm library has been released at <https://git.openi.org.cn/OpenPointCloud>.

CCS CONCEPTS

• **Computing methodologies** → **Neural networks**; • **Software and its engineering** → **Open source model**; • **Theory of computation** → **Data compression**.

KEYWORDS

Point Cloud, Geometry Compression, Open-source Software, Algorithm Library, Deep Learning

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1 INTRODUCTION

In recent years, Point Cloud (PC) has played a more and more important role in 3D representations of scenes or objects, which has promising applications in immersive multimedia, autonomous driving, digital museums, etc. PC is formed by a collection of unstructured points with 3D geometry positional coordinates. Lots of research efforts for point cloud technologies have been carried out, including point cloud compression, processing and analysis [13, 14]. Due to the fast developments of LiDAR and depth cameras, a huge amount of PC data has been produced. Without efficient compression methods, the storage and transmission of PC data will be difficult. Therefore, point cloud compression (PCC) is becoming a very popular research field in both academia and industry.

Moving Picture Experts Group (MPEG) is an international standard organization, which has proposed two popular solutions for PCC: Geometry-based PCC (G-PCC) and Video-based PCC (V-PCC) [7]. In G-PCC, octree-based geometry codec is a commonly used technique since octree-based data structure effectively leverages the sparse structure of PC. Besides, trisoup-based geometry codec, which models the surface of PCs using triangle meshes or triangle soup, is also an important scheme in G-PCC. In V-PCC, PC data is converted into a set of video sequences, which are compressed by video compression methods [12]. Besides the compression solutions provided by the MPEG organization, the deep learning based point cloud compression (DL-PCC) [10] has attracted much more attention, and is expected to achieve higher coding efficiency due to the fast developments of deep learning theories and technologies.

Generally, our work and contributions are summarized as follows: First, we establish the first open-source library of popular

Table 1: A summary of PCC algorithms in our library.

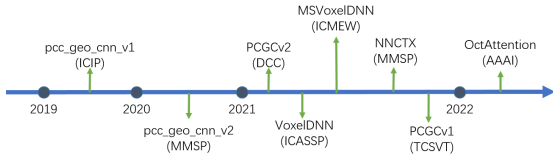
Algorithm	Category	TensorFlow	Pytorch	TensorLayer
pcc_geo_cnn_v1 [10]	lossy geometry	✓	✓	
pcc_geo_cnn_v2 [11]	lossy geometry	✓	✓	
PCGCv1 [18]	lossy geometry	✓	✓	✓
PCGCv2 [17]	lossy geometry		✓	
VoxelDNN [8]	lossless geometry	✓		✓
MSVoxelDNN [9]	lossless geometry		✓	✓
NNCTX [4]	lossless geometry	✓	✓	
OctAttention [3]	lossless geometry	✓	✓	

learning-based PCC algorithms on a public platform. Second, we conduct a series of benchmark tests on different aspects to compare their performances objectively. Third, we provide source codes and pre-trained models in mainstream deep learning frameworks for practical usages, including TensorFlow, Pytorch, and Tensorlayer [5]. Tensorlayer is a multi-backend AI deep learning framework which supports TensorFlow, Pytorch, MindSpore, PaddlePaddle as back ends and makes it convenient to implement the code on different hardware like Huawei-NPU and Nvidia-GPU. Research works over the past few years are summarized to facilitate better understanding and developments of PCC algorithms.

The PCC algorithms contained in our library are listed as follows:

- Four lossy geometry PCC methods including pcc_geo_cnn_v1 [10], pcc_geo_cnn_v2 [11], PCGCv1 [18], PCGCv2 [17]. We will introduce their principles and contributions in Section 2.1. We report comparison results by bits per input point (bpp), point-to-point distances (D1) [16], point-to-plane distances (D2) [15], and time complexity in Section 3.
- Four lossless geometry PCC methods including VoxelDNN [8], MSVoxelDNN [9], NNCTX [4] and OctAttention [3]. General algorithm descriptions are given in Section 2, and evaluation results are demonstrated in Section 3.

Tab. 1 shows the basic information of the PCC algorithm, while Fig. 1 depicts the corresponding timeline of these methods.

**Figure 1: Chronological overview and publishers of the state-of-the-art point cloud compression (PCC) methods.**

2 SUPPORTED ALGORITHM LIBRARY

2.1 Lossy Geometry PCC Methods

The contributions of pcc_geo_cnn_v1¹ are listed as follows:

- It is the first work to use a 3D auto-encoder into PCC.
- It utilizes a loss function that jointly optimizes both compression rate and distortion with a trade-off parameter.
- It combines arithmetic coding with convolutional neural network (CNN) and makes entropy coding applicable.

pcc_geo_cnn_v2² is an improved version of pcc_geo_cnn_v1:

¹https://git.openi.org.cn/OpenPointCloud/pcc_geo_cnn_v1_yh

²https://git.openi.org.cn/OpenPointCloud/pcc_geo_cnn_v2_yehua

- It employs deeper transform modules with more convolutional layers and filters.
- It adjusts the balancing parameter in the focal loss.
- It traverses all numbers from 0 to 1 at a step of 1/256 to get the best threshold value for a specific distortion metric.

PCGCv1³ has a similar network structure to pcc_geo_cnn_v1, but makes improvements on many aspects:

- It uses Voxception-ResNet-based blocks for analysis and synthesis transform to avoid gradient vanishing in training.
- It applies a hyper-prior model as well as a Laplacian distribution function to approximate the probability distribution for arithmetic coding.
- It measures the distortion with a classification-based mechanism named Weighted Binary Cross-Entropy (WBCE).

PCGCv2⁴ is upgraded from PCGCv1 and makes a series of improvements for better coding performance:

- It introduces Inception-Residual Network (IRN) unit to study the geometry features efficiently.
- It uses the multi-scale strategy to fully utilize the geometry features at different scales.
- It applies sparse convolutions to accelerate training and inference period of the network.

2.2 Lossless Geometry PCC Methods

Unlike the methods mentioned above, VoxelDNN⁵ is a lossless PCC algorithm. It provides several innovations as follows:

- It partitions PC data into a set of sub-blocks and uses octree codec to organize the position of sub-blocks.
- It applies a deep generative model which contains masked convolutions to approximate the occupancy probability for arithmetic coding.
- Each block is down-sampled in 3 levels to search for the best down-sampling level by comparing the compression rate.

However, VoxelDNN suffers from high computational complexity. Based on VoxelDNN, MSVoxelDNN⁶ comes out aiming at solving the high time complexity problem. The main features of MSVoxelDNN are summarized as below:

- It leverages multi-scale structures which encodes the sub-blocks in a coarse-to-fine order. Using the multi-scale structures, it avoids high time complexity problem generated by voxel-by-voxel decoding.
- It significantly reduces the time complexity by losing small amount of compression rate.

NNCTX⁷ can be regarded as a practical lossless PCC method. It designs a coding process that realizes efficient decoding and avoids doing it voxel-by-voxel like VoxelDNN. Contributions of NNCTX are summarized as follows:

- It generates 8 candidate voxels from layers in each resolution as extended layers to enrich the context for calculating distribution.

³https://git.openi.org.cn/OpenPointCloud/PCGCv1_yehua

⁴https://git.openi.org.cn/OpenPointCloud/PCGCv2_yh

⁵https://git.openi.org.cn/OpenPointCloud/VoxelDnn_yehua

⁶https://git.openi.org.cn/OpenPointCloud/MSVoxelDNN_yehua

⁷https://git.openi.org.cn/OpenPointCloud/NNCTX_yh

Table 2: Quantitative performance gains of deep learning lossy PCC methods against G-PCC octree. BD-PSNR gains [1] is measured by D1 and D2 distortions. Higher value indicates better performance.

Category	Point Cloud	pcc_geo_cnn_v1		G-PCC trisoup		pcc_geo_cnn_v2		PCGCv1		PCGCv2	
		D1	D2	D1	D2	D1	D2	D1	D2	D1	D2
MVUB	Phil_vox9_0139	1.592	-2.514	2.347	1.531	4.354	1.898	6.912	4.574	10.169	7.425
	Phil_vox10_0139	1.751	-2.282	2.483	1.131	4.945	2.204	7.650	3.448	9.625	6.473
	Sarah_vox9_0023	1.219	-2.592	2.589	1.305	4.320	2.043	6.740	4.703	10.186	7.800
	Sarah_vox10_0023	1.548	-2.324	1.215	1.071	4.987	2.478	5.490	3.670	9.708	6.877
8iVFBv2	Loot_vox10_1200	1.462	-1.931	3.307	1.753	6.542	4.779	8.981	6.948	12.384	10.871
	Queen_vox10_0200	2.031	-1.684	3.012	0.960	7.396	5.158	9.357	6.618	13.317	10.848
	Soldier_vox10_0690	1.796	-1.575	3.601	1.734	6.448	4.691	9.330	7.455	11.908	10.561
	Longdress_vox10_1300	1.795	-1.528	3.619	1.623	6.475	4.688	9.237	7.412	12.276	10.847
	Redandblack_vox10_1550	1.745	-1.617	3.469	1.833	5.988	4.022	9.002	7.115	11.549	10.331
Owlii	Dancer_vox11_0001	2.024	-1.437	3.815	2.999	8.537	6.612	8.569	6.910	12.397	11.238
	Player_vox11_0200	2.083	-1.104	3.728	3.001	8.885	6.933	9.548	7.795	12.431	11.839
Average		1.731	-1.871	3.017	1.722	6.252	4.137	8.265	6.059	11.450	9.555

- It arranges the input context in a specific coordinate by 4 planes, which is an innovation on context modeling.

OctAttention⁸ combines octree and transformer together to predict the distribution for entropy coding. OctAttention encodes the octree occupancy code, and builds a self-attention model to learn the probability distribution of octree code with the context of siblings and ancestors. Besides, OctAttention also proposes a mask operation scheme to encode PC data parallelly, which can accelerate the processing significantly. As octree structure consumes less space than voxelization, the performance of OctAttention is state-of-the-art on compression rate by far.

3 BENCHMARKING RESULTS AND ANALYSIS

3.1 Datasets

To evaluate all these methods thoroughly, we perform tests over PC files from 3 publicly available datasets: 8i voxelized full bodies version 2 (8iVFBv2) [2], Microsoft voxelized upper bodies (MVUB) [6] and Owlii dynamic human mesh [19]. The selected PC datasets cover different bit widths and scales.

3.2 Metrics for Performance Evaluation

For geometry PCC methods, performances are evaluated from several aspects: (1) compression rate, indicated by bpp; (2) distortion, measured by D1 and D2; (3) complexity, evaluated by running time.

3.3 Lossy Geometry PCC Evaluation

For lossy methods, we calculate the Bjøntegaard delta peak signal-to-noise ratio (BD-PSNR) gains to jointly evaluate the performance of bpp and D1, D2 metrics. Results are shown in Tab. 2. PCGCv2 obtains the state-of-the-art performance in open-source lossy PCC methods in all bit widths, benefiting from the network structure with IRN and multi-scale reconstruction structure. The second best is PCGCv1 and the third best is pcc_geo_cnn_v2. pcc_geo_cnn_v1, trisoup, and octree have similar performances.

3.4 Lossless Geometry PCC Evaluation

The principal metric of evaluating lossless PCC methods is bpp as shown in Tab. 3. We can see that the performance of OctAttention is the best, which indicates that the adoption of octree structure is an efficient strategy for point cloud representation. VoxelDNN is the second best since it searches the best path for better compression performance. The third one is NNCTX. Based on VoxelDNN, MSVoxelDNN gets higher bpp with a trade-off of less running time. Overall, it is clear that all deep learning methods perform better than G-PCC octree method, and therefore learning-based PCC methods have a promising future.

3.5 Running Time Evaluation

We draw histogram graphs of running time of all geometry PCC methods in Fig. 2. We calculate the average time of different compression rates. In Fig. 2, the fastest PCC method is G-PCC octree. G-PCC trisoup is also fast as they are not learning-based methods. Although PCGCv2 is a learning-based method, it still ranks the second for that it applies a sparse convolution technique of MinkowskiEngine, which can speed up running significantly. The fourth fastest method is pcc_geo_cnn_v1 which has simple network structure. Although pcc_geo_cnn_v2 has a similar network structure as PCGCv1, it has much higher time complexity as it spends most of the time in searching for an optimal threshold to minimize a given distortion metric. OctAttention has high decoding time, while encoding can be accelerated by parallel processing. VoxelDNN ranks the last mainly because it encodes the PC data voxel-by-voxel.

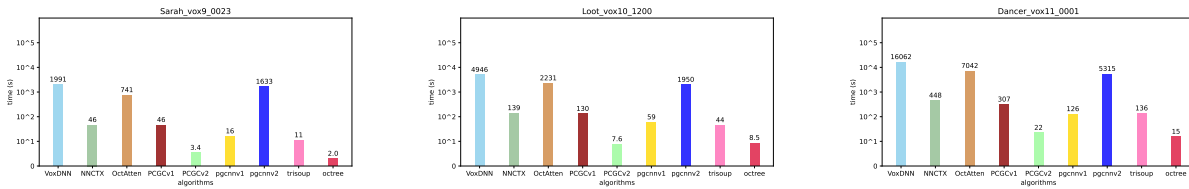
4 CONCLUSION AND FUTURE WORK

In this paper, we make introductions to OpenPointCloud, the first open-source algorithm library in the field of DL-PCC. We have illustrated the principle and contributions of different state-of-the-art PCC methods, and have implemented them on different deep learning programming frameworks, such as TensorFlow, Pytorch and TensorLayer. We have established the performance benchmarking tests for all the point cloud compression methods. The developed open source project and released performance benchmarking results can be expected to boost the advancements of deep learning based point cloud compression. Future work will be extended to

⁸https://git.openi.org.cn/OpenPointCloud/OctAttention_yh

Table 3: Compression performance of all lossless deep learning lossy PCC methods. Compression performances are evaluated in terms of bpp. Lower value represents better performance.

Category	Point Cloud	Point Number	G-PCC Octree	MSVoxelDNN	NNCTX	VoxelDNN	OctAttention
MVUB	Phil_vox9_0139	356258	1.166	1.031	0.997	0.919	0.839
	Phil_vox10_0139	1594740	1.147	1.033	1.021	0.909	0.795
	Sarah_vox9_0023	299363	1.096	0.966	0.929	0.879	0.749
	Sarah_vox10_0023	1339374	1.073	0.969	0.965	0.869	0.727
8iVFBv2	Loot_vox10_1200	805285	0.970	0.797	0.761	0.718	0.628
	Queen_vox10_0200	1000993	0.773	0.822	0.919	0.698	0.671
	Soldier_vox10_0690	1089091	1.030	0.823	0.800	0.739	0.670
	Longdress_vox10_1300	857966	1.015	0.823	0.778	0.738	0.665
	Redandblack_vox10_1550	757691	1.100	0.905	0.885	0.804	0.737
Owlii	Dancer_vox11_0001	2592758	0.881	0.720	0.691	0.661	0.613
	Player_vox11_0200	2925514	0.898	0.716	0.691	0.674	0.631
	Average	1238094	1.013	0.873	0.858	0.783	0.702

**Figure 2: Running time of all methods in different bit width point cloud files. From left to right, the methods are VoxelDNN, NNCTX, OctAttention, PCGCv1, PCGCv2, pcc_geo_cnn_v1, pcc_geo_cnn_v2, G-PCC trisoup, G-PCC octree. The vertical axis represents the total time of encoding and decoding. The left, middle and right represent 9bit, 10bit and 11bit data, respectively.**

include more algorithms of point cloud processing and analysis (i.e., denoising, completion, upsampling, object detection, segmentation and tracking, etc.), to enrich the library of point cloud algorithms.

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